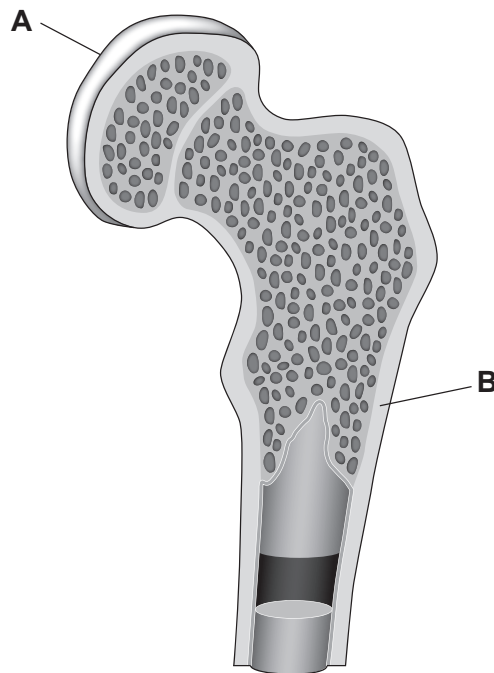


Option B: Human Musculoskeletal Anatomy

7. (a) The diagram below shows a longitudinal section through the head of the femur.



- (i) Name the type of tissue at **A** and the cells which form it. [2]

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- (ii) Bones are continually being remodelled for growth and repair, with calcium ions being both deposited and removed from the bone matrix. What is the name of the bone type **B** and the name of calcium compound found there in the highest proportion? [1]

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- (b) Turners Syndrome (TS) is a condition which occurs in girls who have an incomplete or missing X chromosome. They produce less oestrogen than normal. This has an effect similar to the menopause on bone in reducing the level of calcium compounds, but at a much earlier age if left untreated. Oestrogen at normal levels reduces the number of osteoclasts and their activity.

- (i) Describe the functions of osteoclasts and osteoblasts. [1]

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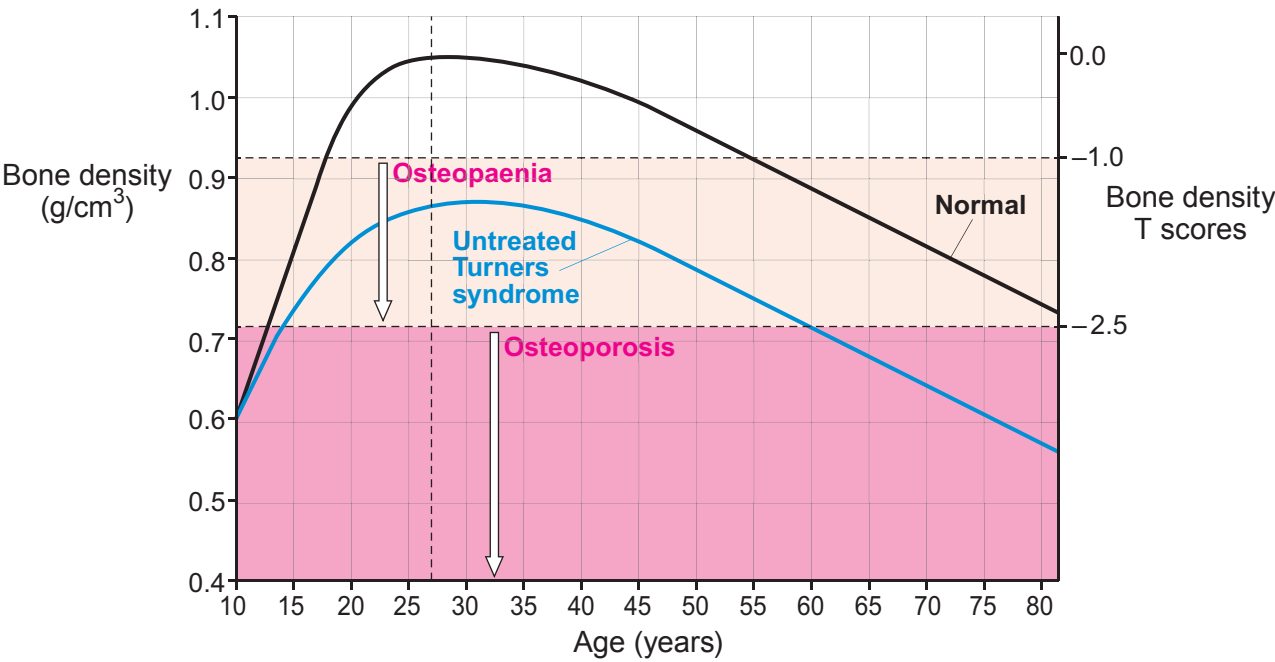
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(ii) Explain why oestrogen treatment might be given to Turners Syndrome patients. [2]

(iii) Turners Syndrome patients have a DEXA scan to assess Bone Mineral Density (BMD). The scan compares density to that of a healthy person of the same age. The difference is then calculated as a standard deviation called a T-score.



A TS patient has a Bone Mineral Density T-score of -2.56 . Suggest a possible treatment for a patient which could be used alongside oestrogen and explain the effect of this treatment. [2]

(iv) With reference to the graph, explain why is it difficult to confidently diagnose a 10-year-old girl with Turners Syndrome using the BMD test alone. [2]



(v) Explain why a standard deviation is used for the BMD scan. [1]

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(vi) It is estimated that there is a 25% increase in the risk of fractures in Turners Syndrome patients, especially of the long bones of the forearm. What would be the course of treatment for a displaced fracture of the tibia? [1]

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(vii) Calcium is needed for the contraction of muscles. Explain how a lack of calcium could cause poor muscle contraction in patients. [4]

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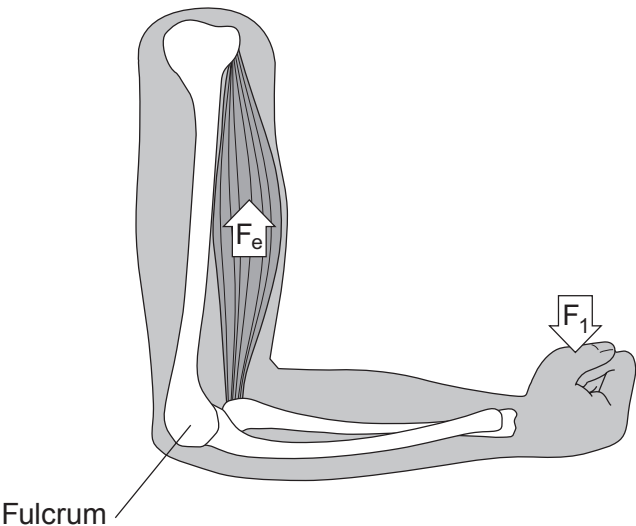
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(c) TS patients have shorter bones than normal, and scientists investigated the effort needed to lift a 4kg hand weight using the biceps muscle.



(i) State the order of lever which is shown in the diagram above. [1]

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- (ii) Calculate the force effort required to lift the 4 kg hand weight in a patient with TS where elbow to hand distance of 34 cm and elbow to biceps is 4 cm. Show your working. [2]

Key
 F_e = force exerted by the effort
 F_1 = mass (kg) $\times$ 9.8 N
 D_1 = distance from load to fulcrum
 D_2 = distance from effort to fulcrum
1 kg = 9.8 Newtons (N)

Using the formula: $F_e = F_1 \times \frac{D_1}{D_2}$

Force effort = N

- (iii) When testing patients, state the factor that would need to be taken into account to make a valid conclusion. [1]

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